

Large language models unlock the ecology of species interactions

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Species interactions shape population dynamics, geographic distributions, evolutionary trajectories, and responses to environmental change. Yet data on these interactions remain scarce across broad spatial, temporal, and taxonomic scales because they are difficult to collect in the field. One promising source of interaction data is citizen science platforms, which contain billions of biodiversity observations, often accompanied by unstructured text comments that may document interactions among organisms. Advances in large language models (LLMs) make it increasingly feasible to identify, extract, and categorize biotic interactions from these unstructured data at scale. Here, we present an LLM workflow that collects species interaction observations from multilingual citizen science comments. Using two case studies—bird–bird and plant–pollinator interactions—we show that LLMs can rapidly extract interaction types and participating species with high accuracy. These data can greatly expand the spatial, temporal, and taxonomic coverage and resolution of species interactions data, enable new tests of long-standing ecological questions, and improve our ability to track ecological changes. With appropriate validation, expert review, and attention to data privacy for both users and sensitive species, this approach opens new opportunities to characterize, forecast, and conserve biodiversity under global change.

citizen science | species interactions | large language model

Species Interactions: A “New Frontier” of Ecological Data

Biotic interactions between species are fundamental to characterizing and predicting ecosystem resilience (1), community dynamics (2), biodiversity patterns (3), ecosystem functioning (4), and evolution (5). Although large-scale data on species abundance and distribution are increasingly available (6), knowledge of species interactions remains limited across many ecosystems worldwide (7, 8). Widespread changes in geographic range and phenological shifts under rapid global change have profound consequences for species, communities, and interactions. For instance, range expansion may introduce new species into communities, creating novel interactions, whereas range contraction can lead to the local extirpation and the loss of existing interactions (9–11). Despite ample case studies, the novel ecological interactions produced by “species on the move” (i.e., shifts in spatial range and phenology) (12) have not been examined on broad

spatial, temporal, and taxonomic scales. Understanding the consequences of these large-scale changes in species interactions is a major frontier in ecology, but progress is constrained by the scarcity of detailed behavioral and interaction data (13).

Such a critical biodiversity data gap persists because species interaction data remain scarce across broad spatial and temporal scales, largely due to the logistical challenges of field-based data collection (7, 14). Traditional efforts rely heavily on expert observations or experiments on a local scale, which are time-consuming and costly, making it challenging to document the full range of interactions that a focal species may experience, leading to spatial, temporal, and taxonomic constraints on resulting datasets (7, 8, 15). Because species interactions are often highly context-dependent (16, 17), small-scale datasets provide only a limited view of ecological communities, constraining our ability to generalize dynamics and predict how interactions may shift under global change.

Here, we propose that citizen science platforms, such as iNaturalist ([inaturalist.org](https://www.inaturalist.org)) and eBird (ebird.org) (18), which collectively contain huge amounts of biodiversity observations, can help alleviate the longstanding scarcity of species interaction data. In addition to manual screening of observation data,

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we highlight the power of automated data collection enabled by advances in large language models (LLMs) through two case studies on bird–bird and plant–pollinator interactions. Finally, we discuss the great potential of this vastly expanded data source to answer long-standing ecological questions and inform biodiversity conservation.

Citizen Science as a Rich Source of Species Interaction Data

Citizen science engages members of the public in scientific data collection, generating vast quantities of ecological observations that increasingly support biodiversity research and conservation science (19, 20). This practice has also been termed “participatory science” (21). In this paper, citizen science specifically refers to volunteer-generated biodiversity data, which is different from community science that traditionally refers to grassroots initiatives aiming to benefit local communities (see refs. 22 and 23 for a more in-depth discussion of the terminology). Observations contributed by participants from around the world have greatly advanced our understanding of biodiversity from species to community levels across diverse groups of taxa and in both populous and remote regions (24–27). At finer scales, data from popular platforms have generated new insights into plant–frugivore (28), plant–pollinator (28–30), and predator–prey interactions (31), and even revealed previously undocumented interactions (32, 33). Beyond individual case studies, the extensive spatial and temporal coverage of citizen science data has enabled comparative analyses of species interactions across analogous ecosystems (e.g., urban ecosystems) (34, 35) and broad environmental gradients (e.g., latitudinal) (36). These data complement observations by trained scientists by expanding the spatial and taxonomic scope of documented interactions and by revealing previously unrecorded interactions, especially in highly biodiverse but often data-deficient areas (e.g., the Neotropics) (28, 30, 31, 33, 35).

Although citizen science has generated valuable data on species interactions, most platforms were originally designed to record the presence and abundance of a single taxon rather than interactions among species. The study of interactions in these datasets has largely relied on the incidental recording of observers either through optional text posts (“comments”), or through uploaded photos or other media types (19). Previous studies used these unstructured data to identify interactions, but the large volume of records has generally limited analyses to manual examination of specific interactions (e.g., plant–pollinator) within relatively small areas (e.g., in one or several metropolitan areas) (28, 29, 31). Efforts are being made to enable the structured submission of species interaction data on some platforms. For example, iNaturalist now provides functionality to explicitly record interactions (37), such as pollination, predation, and parasitism, among others. These records are automatically imported into the Global Biotic Interactions Database (GloBI) (38), where they are made readily accessible for ecological research. Project FeederWatch, a platform designed to record birds in backyard feeders (39), includes standardized submission of bird–bird interactions (displacement or predation) (40–42). Nevertheless, explicit recordings of species interactions remain uncommon across platforms, as most prioritize

simplified reporting protocols that encourage participation and the collection of focal biodiversity data (19).

Despite their promise, current methods of collecting species interaction data from citizen science platforms have yet to realize the full potential of the unprecedented spatial and temporal coverage, resolution, and sheer volume of observations contributed by participants worldwide. For example, we estimate that about 3 to 5% of eBird observations contain comments, and 3 to 5% of comments contain species interactions. With over two billion observations on eBird worldwide (43), these rough estimates suggest there are ~5,000,000 comments that contain species interaction data. Even though information on species interactions is relatively uncommon, because of the sheer volume of comments, they are a data source with great promise.

Automating the Collection of Species Interactions with LLMs

Collecting and analyzing species interaction data from citizen science platforms has become substantially easier due to increasing data accessibility and recent advances in natural language processing, particularly LLMs. Previous bottlenecks in obtaining citizen science observations have been alleviated through centralized biodiversity data infrastructures. For example, the Global Biodiversity Information Facility (GBIF) (44) provides access to observations from multiple platforms, including iNaturalist and [observation.org](https://www.observations.org/). Researchers can query records by species, location, and time period, and filter observations containing comments. These records can be further filtered and processed using LLMs. Depending on the application, LLMs can be trained on manually labeled datasets (e.g., comments that contain species interactions or certain types of interactions) or used in a zero-shot or few-shot framework to identify and classify interactions from unstructured text. Here, we illustrate one potential basic training workflow (Fig. 1) with case studies of bird–bird interactions and plant–pollinator interactions. LLM prompts for each case study are included in [SI Appendix](#).

Case Study 1: eBird Comments. To illustrate the potential of LLMs for extracting species interactions from citizen science platforms, we first fine-tuned a GPT-4o model (version number ft:gpt-4o-2024-08-06; see [SI Appendix](#) for model parameters including bird–bird interaction definitions from the AvianMetaNetwork) (45) on 544 randomly selected eBird comments from the state of Texas (retrieved Aug 2023). The training dataset consisted of an equal number of comments containing species interaction information and comments without interaction information. Within the test set, the fine-tuned model correctly identified 76% of all comments containing interactions and 96% of comments that did not. We then applied the model to eBird observations from Michigan collected between January 2019 and January 2024. Of the 18,150,523 records, 632,169 (3.5%) contained comments, and the model identified 29,345 (4.6%) that contain species interactions. The cross-state application also provided a preliminary test of the model’s ability to generalize to comments originating from a different geographic region than the training data.

To further test the LLM’s accuracy in categorizing species interactions, a single annotator (O.J.S.) manually reviewed and categorized 485 comments randomly selected from those

1. Extract observation comment

Comments can contain vague descriptions of the interaction (A), non-standardized species names (B), and in different languages (C-D).

2. Process with LLM

Automatically extract the information from comments

3. Inspect Manually

Mandatory quality control step to ensure interaction and species names are correct

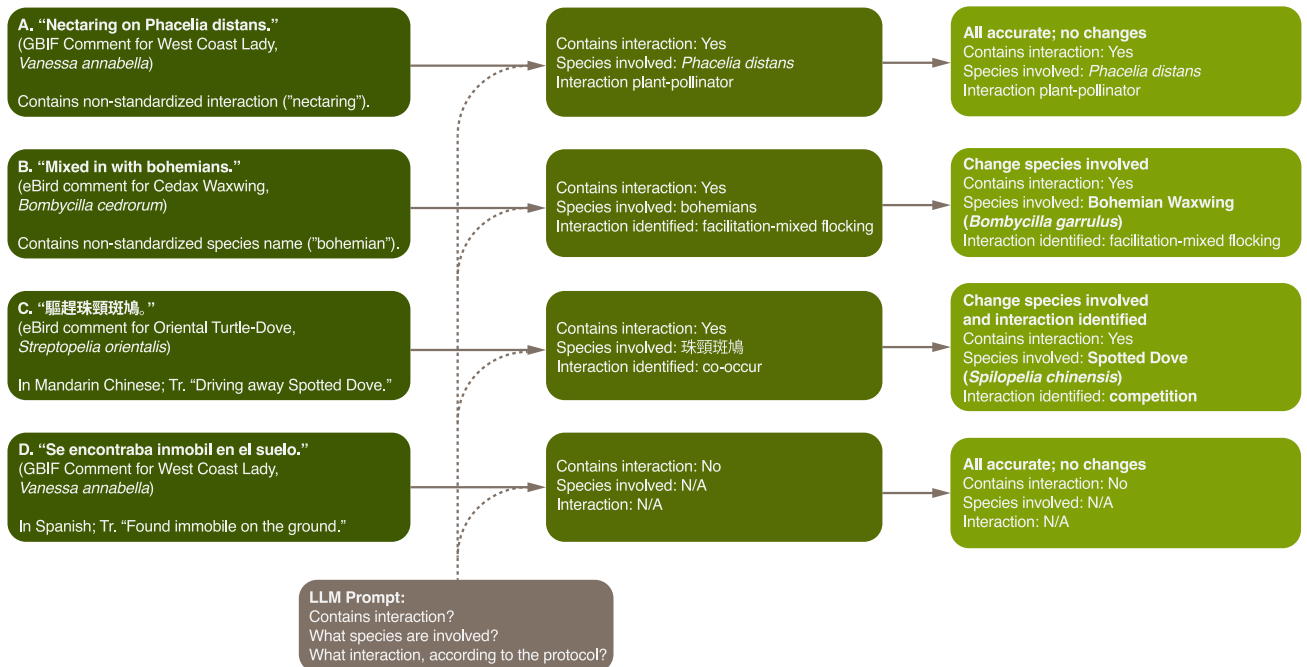


Fig. 1. Basic workflow of extracting species interactions using LLMs. Step 1: Examples of comments from eBird and GBIF. They represent several characteristics of the comments: vague descriptions of interactions, nonstandardized species names, different information density (may or may not contain interactions), and multilingual. Step 2: LLMs can extract the three key types of information in comments of multiple languages: whether a comment contains an interaction, what species were involved in the interaction, and what type of interaction was recorded. To ensure the efficiency of extraction, the nonstandard species names are reported as is. Step 3: Processed comments are then manually inspected to standardize species names according to the specific taxon and to correct any misidentified interaction categories. This step is essential for quality control.

identified as containing interactions in the Michigan dataset. Using the AvianMetaNetwork protocol (45), the annotator categorized interaction types and the species involved, completing the task in approximately 8 h. We then imported the comments into Claude Sonnet 4 (version number claude-sonnet-4-20250514) with a prompt containing the categorization protocol and matched the output to the manually labeled interactions. The LLM was not trained with prior data (zero-shot) and correctly categorized 395 out of 485 (81.44%, Wilson's interval [77.63%, 84.75%]) interactions in less than 5 min. Species were identified by the LLM as the name used in the original comments, which, in many cases, was not the standard common name. For example, the comment "Mixed with bohemians" refers to the species Bohemian Waxwing (*Bombicilla garrulus*) as "bohemian" (Fig. 1, example comment A). After manually correcting these species names to the 2025 version of eBird taxonomy (46), the LLM correctly extracted species identity from 468 (96.49%, [94.33%, 97.88%]) comments, up from 80% before the manual correction, including comments with multispecies interactions (i.e., the focal species interacted with more than one other species).

We then constructed and compared matrices from manually labeled and LLM-extracted interactions and calculated their differences using a series of dissimilarity metrics (47): β_S defines the dissimilarity in species compositions; β_{OS} defines the dissimilarity of interactions between species shared by the two matrices; β_{WN} defines the overall dissimilarity of interactions; β_{ST} defines the dissimilarity of interactions due to turnover of species between the two matrices. Smaller β values indicate higher similarity. For some specific interaction types, such as

mobbing (aggression toward potential predators) and predation, the LLM correctly identified all interactions from the comments (Table 1). For facilitation-feeding (multiple species feeding together) (45), the labeled and extracted matrices based on the presence of species and interactions are identical, but the LLM failed to extract one interaction (Fig. 2). For two common types of species associations, mixed-species flocking (multiple species benefit from being in a group) and co-occurring (spatial and temporal co-occurrence), the manually labeled and LLM-extracted matrices show high similarity (Table 1). Some common interactions are more robust to data extraction because they have been recorded in many comments, while others are more incidental but may indicate rare, understudied cases.

In this case study, we used two LLMs (GPT 4o and Claude Sonnet 4) because the fine-tuning feature, only available for GPT, could help with identifying the small proportion of comments that contain interactions. In later use cases, including case study 2 reported below, we found that Claude Sonnet 4 could achieve comparable performance by analyzing comments directly obtained from eBird without fine-tuning. This indicates that cutting-edge LLMs may be sufficiently capable of identifying species interactions without prior training (zero-shot).

To test the performance of LLMs in relatively data-deficient, non-English-speaking areas, we also analyzed species interactions from 10,000 comments in Taipei, a subtropical city with relatively few eBird comments (62,356 comments) and a lower proportion of comments containing species interactions (1.43% in the sampled subset), with most of the comments in Chinese (see Fig. 1, example comment C; see also SI Appendix for

Table 1. Dissimilarities between interaction matrices constructed from manually labeled interactions vs. LLM-extracted interactions, following (45)

Interaction type	β_s	β_{OS}	β_{WN}	β_{ST}	# Interactions in manually labeled matrices	# Interactions in LLM-extracted matrices
Facilitation-feeding	0	0	0	0	63	62
Facilitation-mixed flocking	0	0.00357	0.00357	0	283	279
Predation	0	0	0	0	13	13
Co-occur	0.0253	0	0.0148	0.0148	103	100
Mobbing	0	0	0	0	21	21
Amensalism	0	0	0	0	8	8
Competition	0	0	0	0	3	3
Aggression-potential competition	0	0	0	0	3	3
Aggression	0	0	0	0	6	6

detailed methods). The LLM successfully identified all but four comments not containing interactions (false positive rate 0.04%). Among the 143 comments with interactions, 20 were identified as no interactions (a false negative rate of 14.00%), and an additional 35 comments were identified as different interactions, leading to an overall accuracy of 61.54% (88/143, Wilson's interval [53.00%, 69.44%]). Examining the errors, the model appeared to perform poorly in identifying bird species

in Chinese, and the lack of bird species in comments led to the model concluding no interactions were recorded.

Case Study 2: Comments Regarding a Single Pollinator from GBIF. To illustrate how LLMs can help collect plant-pollinator interactions from citizen science platforms, we downloaded all human observations of a single pollinator, the West Coast Lady (*Vanessa annabella*), from the Global

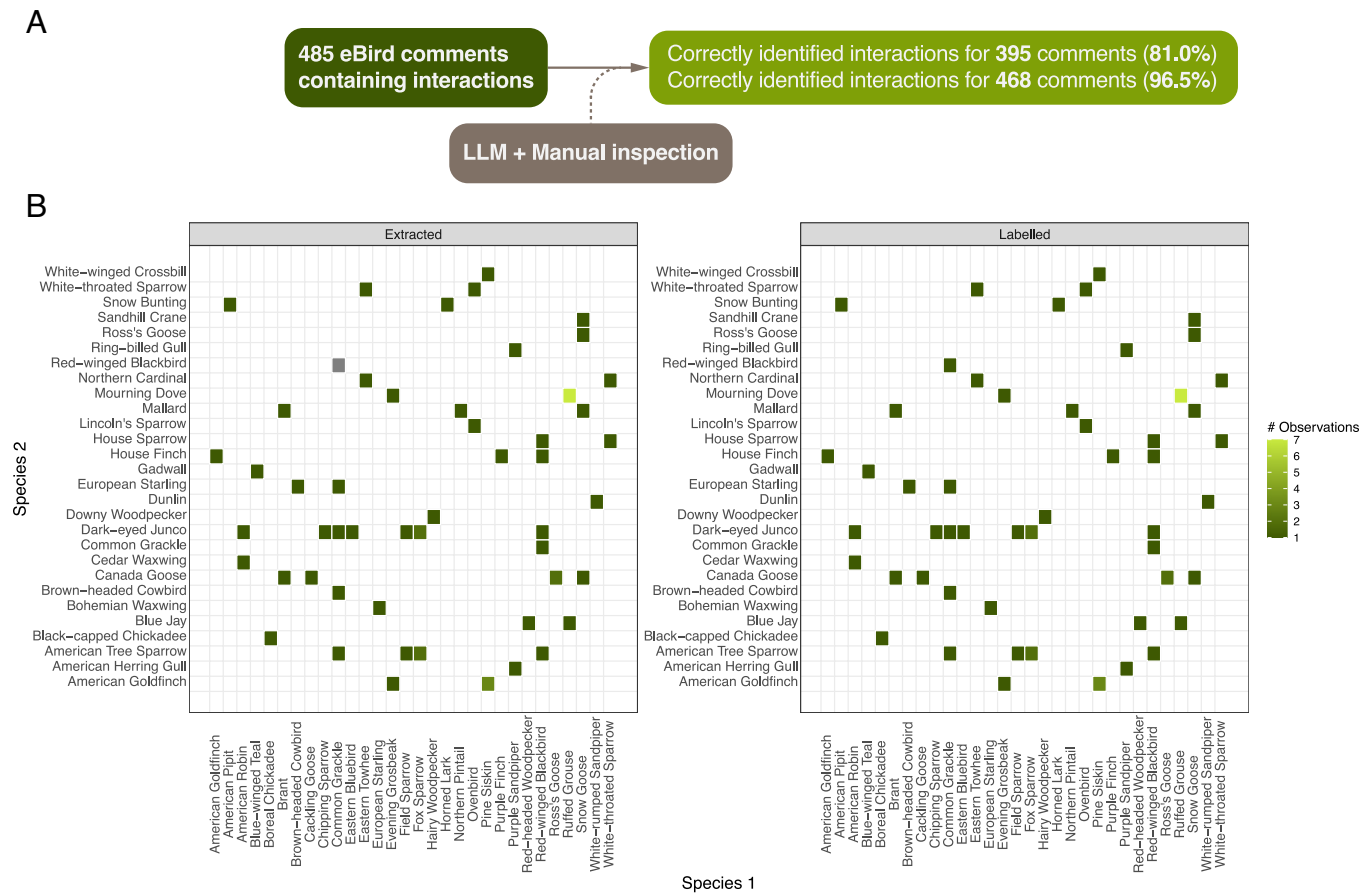


Fig. 2. Interaction matrices of bird species feeding together in Michigan, winter 2023 and 2024. Panel (A) shows the analysis pipeline. Panel (B) Left: the interaction matrix extracted from the LLM; panel (B) Right: the interaction matrix from comments that were manually categorized according to the same protocol. The two matrices are extremely similar, and the LLM failed to extract only one interaction from a comment with Common Grackle (*Quiscalus quiscula*) as the focal species and mentioned Red-winged Blackbird (*Agelaius phoeniceus*) as a participating species (gray square on the Left panel). However, this difference did not reflect on the structure of interaction matrices based only on species identity and interaction presence (i.e., nondirectional, unweighted matrices; see Table 1 for quantitative comparison) because the interaction between the two species was identified in other comments. Overall, this result indicates that the LLM can accurately extract interactions between birds from eBird comments.

Biodiversity Information Facility (GBIF) on Apr 25, 2025. Out of the 8,999 observations, 996 (11.07%) contained comments in both English (934 comments) and Spanish (62 comments; see Fig. 1, example comment D). First, we tested the accuracy of LLMs in categorizing interactions. We manually annotated 400 comments, categorizing all interactions involving the focal species as pollination, herbivory, co-occurrence, and no interactions (SI Appendix). Using the same protocol, we constructed a prompt for Claude Sonnet 4, then asked the LLM to categorize the interactions and extract the other species involved in the interaction. The LLM was not trained on prior data. Using the manually annotated data as validation data, we found that the LLM correctly identified 368 out of 400 (92.00%, Wilson's interval [88.78%, 94.38%]) of all interaction types, and 396 out of 400 (99.00%, [97.28%, 99.68%]) of all species involved after correcting for nonstandard species names (common names, spelling errors, etc.). This subset contained 18 comments in Spanish, and we found no sign of difference in accuracy for identifying interaction types or species involved (Chi-square test, both $df = 1$, $P > 0.80$), although the sample sizes were too small to systematically test for the potential difference among languages.

Then, we used the LLM to extract and categorize interactions from all 996 comments, yielding 299 (30.02%) comments that

contained interactions, 100 of which contained observations of pollination. To compare the completeness of this plant-pollinator dataset, we downloaded all interactions involving *V. annabella* visiting flowers from the Global Biotic Interactions Database (GloBI) (38); the GloBI subset included interactions between *V. annabella* and 138 unique plant species. We converted the common names of those plants in comments to their scientific names (Fig. 3), then aligned them to the GBIF backbone taxonomy (48) used by GloBI. We found 46 species matching the GloBI species list, 36 taxa that matched the GloBI species list at a genus level (i.e., different species in the same genus, or species only identified to genus; hereafter “genus-match”), and 20 taxa that did not match any taxa in the GloBI species list (hereafter “no-match”; Fig. 3).

Although the frequency of reports in this dataset does not necessarily reflect the actual frequency of visitation by the butterfly, the presence/absence of specific species or groups can still reveal important ecological insights. From these results, we made three ecologically important observations. First, among the top 10 species with the greatest number of observations, only three are native to the range of the butterfly (western United States and Mexico). Exotic species are even more prevalent among no-matches, with only six native taxa on the list. The abundance of exotic species on this list corresponds to the global pattern of introduced species integration into

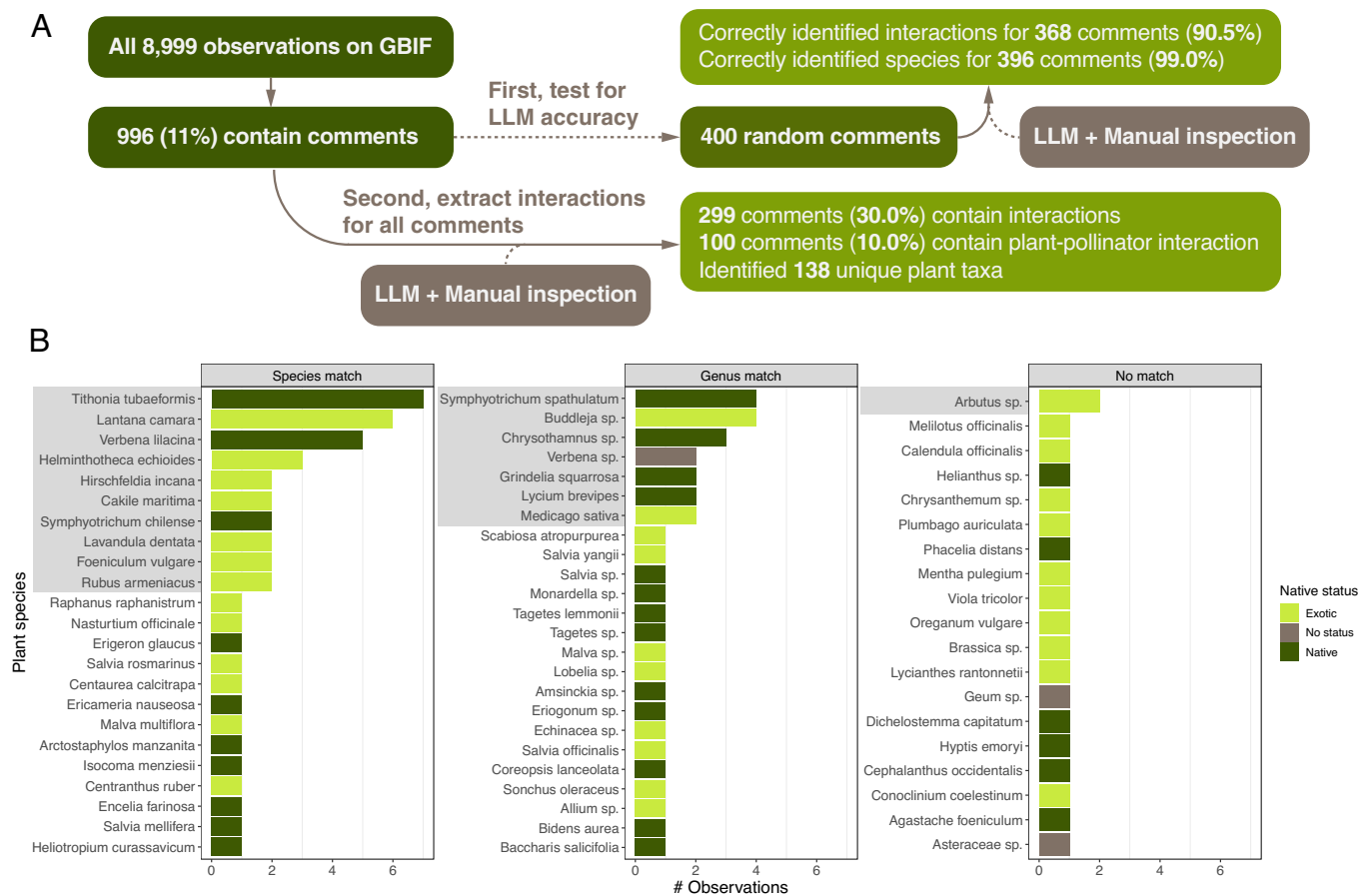


Fig. 3. Plant-pollinator interactions of the West Coast Lady butterfly (*V. annabella*) collected from comments on citizen science platforms. Panel (A) shows the analysis pipeline. In panel (B), the x-axis shows the number of comments observing the interaction. The three columns indicate whether the species also appears in the GloBI species list for the plants visited by *V. annabella*: “species match” means that the plant species in comments exactly matches a species in GloBI, “genus match” means that the plant species matches to the genus level (different species in the same genus, or species only identified to genus), and “no match” means that the plant species is not in the GloBI species list. Species are colored by their native status in the butterfly’s range (western United States and Mexico); “no status” indicates the identified taxon is too broad for determining native status. Species shaded in gray appear in more than one observation, indicating their potential ecological importance.

plant–pollinator networks (49), and the high proportion of exotics among no-matches indicates that observations on GloBI may have overlooked or underestimated introduced plant species used by the butterfly (i.e., “novel” plant–pollinator interactions that can potentially rewire the interaction networks). Second, we observed genus-matches of common ornamental plants, such as butterfly bush (*Buddleja* sp.) and verbena (*Verbena* sp.), likely because observers are often unable to determine species-level scientific names of these ornamental cultivars. These observations highlight the spatial bias of GBIF data toward residential areas in urban and suburban landscapes. Third, the genus-match with the most observations, western mountain aster (*Symphylotrichum spathulatum*), is a native species with a congener (Pacific aster [*Symphylotrichum chilense*]) that is among the top 10 of the exact matches. This indicates that data from citizen science observations may have recorded an important native plant associated with this butterfly absent in GloBI. Overall, this case study highlights that GloBI data, compiled from a wide range of studies and sources and often considered a more complete data source of interactions, can still be improved by the additional data collected from GBIF comments. Furthermore, the gap between GloBI and data extracted from GBIF comments may indicate important gaps in ecological knowledge.

Advantages, Limitations, and Recommendations

In both case studies, LLMs were remarkably fast, accurate, and had high batch-processing ability (processing 500 comments in less than 5 min). Significantly, a further advantage of LLMs is their ability to handle comments in different languages in both of our case studies (Fig. 1). Because citizen science platforms are widely used around the world, the ability to automatically process multiple languages combats traditional biases against non-English data sources and opens broader possibilities for the research of species interactions globally (50). Compared to other AI-enabled methods, such as extracting species interactions from computer vision (51), analyzing text using LLMs has lower computational costs, requires less domain knowledge in AI, and draws information from a large amount of data—writing textual comments is less cumbersome than taking and uploading photos to some citizen science platforms. The latter is especially relevant for eBird since research-grade bird photos often require high-end cameras inaccessible to many observers. While LLMs can also extract species interactions from published scientific papers (15), compared to the wide spatial, temporal, and taxonomic coverage from millions of users around the world, scientific literature is more subject to the same data shortfall of previous datasets on species interactions (7, 8) and may not fully represent the spatial and temporal turnover of interactions. Together, the LLM-based analysis of citizen science comments has the unique potential to revolutionize the collection of interaction data, greatly advancing research on topics from classic ecological questions to emergent issues under global change.

Through the case studies, we identify three major limitations of LLMs. First, results from the model may not be consistent among different runs (e.g., one comment could be classified

as having different types of interactions by the same model) (52). This can be improved by training the model with manually labeled data to improve accuracy, running the model several times, crafting unambiguous prompts with direct instructions, and, most importantly, manually inspecting the model output. Second, LLMs sometimes cannot categorize the interactions or infer standard common names of participating species mentioned in comments, which often contain vague language to describe interactions or when referring to species (Fig. 1). This can be improved by providing more specific prompts that are supplied to the model (e.g., listing keywords that may appear in comments that describe certain interactions or specific examples; SI Appendix), manual inspection, or cross-checking with videos or photos associated with the same observation, either manually or by using deep learning models specializing in computer vision (53). Third, our case studies revealed considerable differences in the accuracy of LLMs among comments in English and Chinese, but not among English and Spanish. This difference in performance among human languages is especially pertinent because data-deficient regions for species interactions often also use languages that are not well-parsed by major LLMs (50, 54). In our case study on eBird comments from Taipei, the low performance might be due to a lack of training text on bird names in Chinese. This gap can be partially mitigated by fine-tuning the model in collaboration with local experts who use the language, which can also in part address the global imbalance in biodiversity data and ecological research (55).

We also flag two common issues of LLMs that are important to consider in other use cases. First, LLMs can generate content that is absent in the input, a common problem called hallucination (56). We did not find any obvious cases of hallucination in our case studies, partly because we gave detailed instructions in our prompt to extract information from short input (comments) into a short, standardized format (either “Yes” or “No” for identifying species interactions, or a set of choices for categorizing species interactions). This rigorously formatted data extraction is less prone to hallucination, which usually arises from longer content generation. Second, output from LLMs can be highly sensitive to the wording of the prompt (57). In our case studies, we did not find large differences in results from different iterations of the prompt during testing, partly due to the specific categorization protocols that include common keywords indicating certain types of interactions and the standardized output format.

In our case studies, we used closed-source models such as GPT 4o and Claude Sonnet 4. Compared to open-source models, these models may be subject to changes without notice, limiting the reproducibility of analyses. The gold standard of scientific research is reproducibility, and to achieve this, we encourage researchers to use open-source LLMs configured locally (offline), allowing for better customization, higher consistency, and lower resource requirements that reduce environmental impacts. We tested the performance of two open-source models, LLaMA 3 8B and Mistral 7B v0.2, and found that using the same test dataset (eBird comments used in case study 1), the accuracy of categorizing species interactions was much lower even after fine-tuning (55.3% and 42.8%; see SI Appendix for details). Although open-source models produced less accurate results at the time of this test (May 2025),

with recent technological advances, we encourage researchers to explore open-source options whenever possible.

In addition to general guidelines for using AI tools in ecological research (14, 58, 59), we provide the following recommendations for obtaining, extracting, and curating species interaction data from citizen science comments. Citizen science data have spatial, temporal, and taxonomic biases and are inherently subject to observation errors (60–62). One specific limitation is the bias toward easily observable, “charismatic” taxa in citizen science datasets, such as the birds and butterflies covered in our case studies. Measures should be taken to address the widespread detection biases and errors of species interactions in citizen science data (e.g., higher detection for more common and more conspicuous species and their interactions, or misidentification of species interactions) using statistical methods that explicitly correct for these biases (63, 64). When obtaining the data, we recommend setting clear research objectives and questions limited to specific species, interactions, or spatial and temporal extents. Clear objectives will help design specific prompts that improve accuracy and limit the amount of data needed to be processed by LLMs and manually, which ensures the benefit of large-scale data analysis with a reasonable computational cost and workload. When extracting data, we recommend clear, detailed prompts with rigorous biological definitions of targeted interactions and short, structured output to reduce hallucination and variability in LLMs. This step is key to the performance of data extraction and requires customization and ample testing, ideally with a manually labeled dataset as illustrated in both case studies above. When curating and publishing data, researchers should adhere to the terms and conditions of specific citizen science platforms and implement measures to protect user privacy and sensitive species. For instance, eBird and iNaturalist terms do not explicitly prohibit the reuse of materials for AI tools (65, 66) and will hide coordinates of sensitive species unless requested (67, 68). Other measures of privacy protection, such as obscuring exact coordinates by a small buffer and redacting names of observers and specific locations, should be implemented when appropriate. Finally, we recommend that all output should be reviewed by experts with domain knowledge to ensure the accuracy of extracted data. While part of the advantage of this approach is the possibility of revealing previously unrecorded interactions, as showcased by case study 2, the possibility of these interactions should be checked by domain experts before further analysis.

Outlook: What Can Large-Scale Interaction Data Tell Us?

Here, we outline several research themes that can be greatly accelerated by extracting large-scale data on species interactions from citizen science data using LLMs.

Biogeographical Patterns of Species Interactions. Many classic themes of biogeography center on species interactions, such as the role of biotic interactions in determining species ranges (69), and latitudinal and elevational gradients of interaction strengths (70). Past studies that have characterized these macroecological patterns relied on datasets collected from a few representative sites and focused on a specific interaction. For example, latitudinal patterns of seed predation and insect predation can be derived from spatially replicated distributed

experiments, but they often lack taxonomic information of predators (70, 71). Other studies have inferred the role of species interactions, such as competition, in shaping geographical ranges across latitudinal or elevational ranges using observed and modeled species distributions (69, 72). While these approaches have generated valuable insights into the role of species interactions in shaping biogeographic patterns, they lack spatial and taxonomic coverage, diversity of interaction types, identity of specific species involved, or direct evidence of species interactions. After correcting for spatiotemporal biases, data collected from citizen science platforms could address these shortfalls by providing high-resolution, human-observed interaction data across large spatial scales and over time. In addition, these data can also advance our understanding of how interactions are likely to shift over time with global change (73), for example, with geographical range shifts of participating species (74), or along urban–rural gradients (75). Together, these large-scale species interaction data pave the first steps into building a “biogeography of species interactions” (76).

Structure and Turnover of Interaction Networks. Interaction networks describe how species in diverse communities are connected via different types of interactions, and these networks are essential to understanding the energy flow and stability of communities (77). Due to logistical constraints, interaction networks constructed using expert observations can often only focus on spatial coverage, temporal coverage, or the resolution of species interactions. For example, observations of plant–pollinator interactions over time provide detailed information on their temporal turnover (78) but are limited in both taxonomy (e.g., only including flowering plants and their pollinators) and space (data only taken from a few dedicated sites). On the other hand, a few highly resolved interaction networks aim for complete coverage of focal taxa and their interactions in single communities (77), but they lack coverage in both space and time. As the strengths and directions of species interactions can change across space (73), time (79–81), and environmental conditions (82, 83), these fragmented networks can only provide incomplete pictures of interaction networks in nature (84). With broader spatial, temporal, and taxonomic coverage of species interaction data, it would be possible to expand both the scope and resolution of interaction networks greatly. This expansion would allow for analyses of the turnover of interaction networks (78, 85, 86), evaluating the effect of time, space, and global change factors in “rewiring” biotic interactions and shaping ecological communities.

Conservation and Discovery of Biodiversity. Natural history observations spur discoveries in ecology and can promote more effective conservation. With increasing coverage across space, time, and taxonomy, citizen science platforms can expand natural system monitoring programs (19). For example, there is little overlap between plant–hummingbird interactions collected by experts and those recorded on citizen science platforms in Brazil (29), indicating the latter’s great potential in revealing previously unrecorded interactions even among well-studied taxa. Examinations of citizen science data have also led to the discovery of cockroaches as pollinators in Spain (32) and the previously

undocumented nectar-robbing behavior of a hummingbird in Bolivia (33). These discoveries can facilitate the description and conservation of biodiversity and are especially important in diverse but data-deficient areas such as the tropics. Data from citizen science platforms can also assist conservation practices. For example, some comments may report information about banded individuals or individuals with tracking devices, which is useful for tracking the focal animal's location, condition, and biotic interactions (87). Furthermore, novel interactions arise when invasive alien species expand into new habitats. Citizen science has long been useful for monitoring the spread and consequences of invasive alien species (88, 89), and these numerous observations likely contain documentation of their interactions with native species that can be used to inform invasive alien species management.

Support for Other Existing Tools and Data. Recent progress in statistical modeling and nongenerative machine learning has provided some solutions for ecologists to overcome species interaction data limitations. Large-scale interaction data collected from citizen science platforms can further expand and support these tools. Statistical models such as joint species distribution models enable the extraction of covariances, or even interaction strengths, from time series data of species abundances across space (82, 90). However, these statistical associations between species abundances could arise from many mechanisms other than interactions and thus do not reflect true underlying interactions (91, 92). Data on recorded species interactions can be used to inform such model-fitting efforts: If two species are known to compete from data records, then the negative associations between their abundance or distribution may be a consequence of this negative interaction. Nongenerative machine learning models, such as random forest and deep learning models, can predict species interactions from existing interaction networks and other data that may influence the potential of species interactions: For example, species with certain matching functional traits are more likely to interact (8, 93, 94). However, these machine learning models still require a large amount of training data and are therefore limited by the low availability of species interaction datasets. Large-scale interaction data from citizen science platforms can provide more data for the training process, improving the accuracy of such predictions (8).

Outstanding Questions

Here, we provide several outstanding questions that are relevant to both ecologists and data scientists, from broad to focused. We hope these questions can spur new and exciting research using this novel approach.

1. How can data collected from comments by LLMs address the rapid "rewiring" of species interaction networks in regions experiencing profound environmental changes?
2. How do the frequency, diversity, and structure of species interactions vary across urbanization gradients, and can citizen science platforms uncover consistent patterns across cities worldwide?
3. Can species interaction observations extracted from citizen science comments improve our understanding

of impacts from disturbances, such as wildfire, drought, habitat fragmentation, or species invasions?

4. Which species groups, geographic regions, habitat types, and interaction types have less available data and thus have the potential to benefit greatly from the large amount of data collected by LLMs?
5. How can the widespread sampling bias in citizen science (spatial, temporal, geographic, taxonomic) be addressed more effectively when collecting interaction data?
6. To what extent do introduced species integrate into existing ecological networks, and are citizen science comments disproportionately capturing novel interactions involving invasive or ornamental species?
7. How can data extracted from comments help understand the interaction between humans and wildlife, both in terms of human-wildlife interaction and conflict, and human perception of nature?
8. What types of species interactions are most easily identifiable using LLMs, what types of interactions require much more nuance, and how do we address this difference in detectability among interactions?
9. How can LLMs be integrated into other advanced AI tools on topics of ecology, such as computer vision models and deep learning models that can analyze photos and videos?
10. What other fields and topics in ecology, behavior, and conservation can be addressed by the increasingly available species interaction data?

Conclusion

Around the world, tens of thousands of observations are uploaded to citizen science platforms every hour. Data contributed to these platforms have greatly increased our understanding of species abundances and distributions. The application of LLMs to identify and analyze species interactions from submission comments has the potential to further unlock the power of citizen science data in describing and understanding species interactions, a key component in describing and predicting biodiversity. With high spatiotemporal resolution and coverage, the immense amount of data offers a great opportunity for research on classic and emerging ecological questions, ample support for existing and emerging tools of ecological analyses, and a source to inform conservation practices under a rapidly changing global environment. We believe that responsibly leveraging LLMs to collect species interaction data is a key example of how advanced AI can be used to accelerate the fields of ecology and global change biology.

AI Statement. We used large language models (GPT 4o and Claude 4 Sonnet) to extract information about species interactions from user-submitted comments on citizen science platforms. We included detailed descriptions on how the models were used in our paper, along with the prompts and version numbers. We did not use generative AI software in any other aspect of this paper's production, including but not limited to writing, visualization, and data analysis.

Data, Materials, and Software Availability. Original data for case studies were obtained from eBird (ebird.org) and Global Biodiversity Information Facility (GBIF; gbif.org). Data processed from the large language model and code used

for the analyses are available at Figshare (95). Prompts containing categorization protocols for species interactions are included in [SI Appendix](#).

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